

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1030

Clover 345 kV
September 27, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1030. The Interconnection Customer proposes a 781.8 MW natural gas combined cycle facility interconnecting at the Clover 345 kV substation in the Virginia area. The request was received on July 18, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 254 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$44,433,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HUTCHINS-COVER 138.0 kV	CLOVER-BEAR 345.0 kV	114.7	464	14.6
CLOVER 345 kV/138 kV XFMR	BEAR SUBSTATION-CLOVER 345.0 kV Line	107.6	495	14.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HUTCHINS-COVER 138.0 kV	Rebuild line with higher-capacity conductor	\$29,011,000	100% (\$29,011,000)
CLOVR 345/138kV XFMR	Replace transformer with higher-rated unit	\$15,422,000	100% (\$15,422,000)
	Total allocated to project		\$44,433,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.